Two matroids can have common chromatic number two and list chromatic number three

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Editorial note. Complete original author construction, definitions and supporting context retained; original list_coloring.bib used. No independent counterexample search, proof repair, added assumptions or mathematical correctness certification.

Editorial note. Mechanical source0.tex adapter excludes the unavailable titling package; no title/author macros from this package are used in the standalone body.

Allowed standard prerequisites (editorial).

- Graphic and partition matroids and list edge coloring. Native definitions and cited standard matroid reduction/Galvin results retained with original bibliography.. The specific coupled three-copy construction and its list obstruction remain inside the counted proof.

1 Introduction

Let $M = (E, \mathcal{I})$ be a loopless matroid. A *coloring* of M is a map $c\colon E \to C$ such that every color class $c^{-1}(\gamma)$ is independent in M. The *chromatic number* $\chi(M)$ is the minimum number of colors in such a coloring, or equivalently the minimum number of independent sets whose union is E. By the matroid partition theorem of Edmonds [8], an optimal coloring can be found in polynomial time in the independence-oracle model; see also [14, Chapter 42].

Now let $M_1 = (E, \mathcal{I}_1)$ and $M_2 = (E, \mathcal{I}_2)$ be two matroids on the same ground set. A coloring of (M_1, M_2) is a coloring of E in which every color class is independent in both matroids. We call the minimum number of colors in such a coloring the *common chromatic number* and denote it by $\chi(M_1, M_2)$. In contrast to the single-matroid case, even deciding whether two colors suffice is hard in general; see [3].

List coloring asks whether a coloring can be chosen when the available colors depend on the element. A *list assignment* is a family $L = \{L(e)\colon e \in E\}$ of finite sets of colors. An *L-coloring* of a matroid is a coloring c satisfying $c(e) \in L(e)$ for every $e \in E$. The *list chromatic number* $\chi_\ell(M)$ is the least integer k such that an L-coloring exists for every list assignment with $|L(e)| = k$ for all e. Seymour proved that $\chi_\ell(M) = \chi(M)$ for every loopless matroid [15].

The same definition applies to a pair of matroids: an L-coloring of (M_1, M_2) assigns to each element a color from its list so that every monochromatic set is independent in both matroids. We call the corresponding parameter the *common list chromatic number* and denote it by $\chi_\ell(M_1, M_2)$. Galvin's theorem on the list chromatic index of bipartite multigraphs gives an important special case [9]. Indeed, the matchings of a bipartite multigraph are precisely the common independent sets of the two unit-capacity partition matroids defined by its two vertex classes. Hence $\chi_\ell(M_1, M_2) = \chi(M_1, M_2)$ whenever both matroids are unit-capacity partition matroids.

Király and Pap studied the corresponding *joint list-coloring property* [11]. Király later asked whether $\chi_\ell(M_1, M_2) = \chi(M_1, M_2)$ for every pair of matroids [10]; the same question was considered by Bérczi, Schwarcz, and Yamaguchi [4] and was later restated as a conjecture by Aharoni, Berger, Guo, and Kotlar [2, Conjecture 5.4]. For three matroids the equality was already known to fail: Király gave an example of three partition matroids on six elements with common chromatic number two that are not list colorable from arbitrary two-element lists; see [4].

1.1 Previous work

For a single matroid, Seymour's theorem [15] was strengthened by Lasoń [12], who allowed the prescribed list size to depend on the element, and by Lasoń and Lubawski [13], who proved the corresponding on-line statement.

For intersections, Király and Pap [11] proved the joint list-coloring property for two transversal matroids, for two rank-two matroids, and in an arborescence setting. Aharoni and Berger [1] proved an ordinary $(p+q)$-coloring bound when the two matroids are respectively p- and q-colorable and p divides q; Berger and Guo [5] removed the divisibility assumption and proved the stronger $(p+q)$-list-coloring statement. Bérczi, Schwarcz, and Yamaguchi [4] developed reductions to partition matroids and obtained list-coloring bounds for transversal, graphic, paving, and gammoid matroids. Aharoni, Berger, Guo, and Kotlar [2] proved in particular that $\chi_\ell(M_1, M_2) \le 2\chi(M_1, M_2)$ for every pair of matroids.

1.2 Our results

Our main result shows that the common chromatic and common list chromatic numbers of two matroids need not be equal.

Theorem 1.1. *There exist a graphic matroid M_1 and a partition matroid M_2 on the same ground set such that $\chi(M_1, M_2) = 2$ and $\chi_\ell(M_1, M_2) = 3$.*

This gives a counterexample to the conjecture mentioned above. The construction is rather simple: it consists of three copies of K_4 in the graphic matroid and a partition of their 18 edges into pairs in the second matroid. The non-2-list-colorability has a direct proof; the upper bound $\chi_\ell(M_1, M_2) \le 3$ follows from the partition-matroid reduction for graphic matroids in [4] and Galvin's theorem.

We also show that if M_1 and M_2 are strongly base-orderable and $\chi(M_1), \chi(M_2) \le 2$, then $\chi_\ell(M_1, M_2) \le 2$. By a theorem of Davies and McDiarmid, the hypothesis is equivalent in this class to $\chi(M_1, M_2) \le 2$.

1.3 Preliminaries

All matroids in the paper are finite. We use standard matroid notation; see for example [14]. A *unit-capacity partition matroid* is specified by a partition $E = P_1 \cup \cdots \cup P_m$; its independent sets are the subsets meeting each P_i in at most one element.

If $M = (E, \mathcal{I})$ and $N = (E, \mathcal{J})$ are matroids on the same ground set with $\mathcal{J} \subseteq \mathcal{I}$, we say that N is a *reduction* of M. Thus every independent set of N is independent in M. Reductions to partition matroids were used in [4] to obtain bounds on list coloring of matroid intersections.

We also recall strong base orderability. A matroid M is *strongly base-orderable* if for every pair of bases B_1, B_2 there is a bijection $\varphi\colon B_1 \to B_2$ such that $(B_1 \setminus X) \cup \varphi(X)$ is a basis for every $X \subseteq B_1$. Strongly base-orderable matroids include all transversal matroids, and hence all partition matroids, and the class is closed under taking minors; see [14, Section 42.6c]. Davies and McDiarmid proved that if M_1 and M_2 are strongly base-orderable, then $\chi(M_1, M_2) = \max\{\chi(M_1), \chi(M_2)\}$ [7].

2 List coloring of two matroids

2.1 Separation of chromatic and list chromatic numbers

Proof of Theorem 1.1. Let M_1 be the direct sum of three copies of the cycle matroid of K_4. Let E_i denote the edge set of the ith copy. In each copy, the six edges are partitioned into the three pairs of opposite edges, which we denote by $\{a_i^+, a_i^-\}$, $\{b_i^+, b_i^-\}$, and $\{c_i^+, c_i^-\}$ for $i = 1, 2, 3$. We use the following elementary fact about K_4.

Claim 2.1. *Two complementary 3-edge sets of K_4 are both spanning trees if and only if exactly one pair of opposite edges is split between them.*

Proof. If an opposite pair is contained in one part, then cardinality forces another opposite pair to be contained in the other part and the third pair to be split. Both parts are then connected and have three edges, hence are spanning trees. Conversely, suppose that no opposite pair is contained in either part. Then all three opposite pairs are split. A set obtained by choosing one edge from each opposite pair of

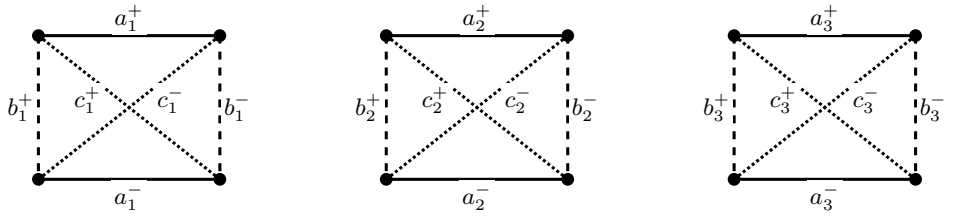


Figure 1: The three copies E_1, E_2, E_3 of K_4 defining M_1. Solid, dashed, and dotted edges form the a-, b-, and c-pairs, respectively.

K_4 is either a triangle or a star, and its complement is respectively a star or a triangle. Thus the two sets cannot both be spanning trees. □

Let M_2 be the unit-capacity partition matroid whose classes are

$$\{a_1^+, a_1^-\},\ \{a_2^+, a_2^-\},\ \{a_3^+, a_3^-\},\ \{b_1^+, c_2^-\},\ \{b_2^+, c_3^-\},\ \{b_3^+, c_1^-\},\ \{c_1^+, b_2^-\},\ \{c_2^+, b_3^-\},\ \{c_3^+, b_1^-\}.$$

We first show that $\chi(M_1, M_2) = 2$. Let

$$X = \bigcup_{i=1}^{3} \{a_i^+, b_i^+, b_i^-\} \quad \text{and} \quad Y = \bigcup_{i=1}^{3} \{a_i^-, c_i^+, c_i^-\}.$$

In each copy of K_4, the three edges of X form a spanning tree, and the same is true for Y. Hence X and Y are independent in M_1. Every class of M_2 has one element in X and one in Y, so both sets are independent in M_2 as well. Thus $\chi(M_1, M_2) \le 2$. Since the entire ground set is dependent in M_1, equality holds.

We next give a two-element list assignment L for which no L-coloring of (M_1, M_2) exists. Give every edge in the first, second, and third copies of K_4 the lists $\{1,2\}$, $\{2,3\}$, and $\{1,3\}$, respectively; see Figure 1. Suppose that an L-coloring exists. In each copy of K_4, only two colors are available. Each color class is a forest and therefore contains at most three edges. Since all six edges are colored, each of the two color classes has exactly three edges and hence is a spanning tree of that copy of K_4. By Claim 2.1, exactly one of the three opposite pairs is bichromatic.

The pair $\{a_i^+, a_i^-\}$ is a class of M_2, so its two edges receive different colors. It is therefore the unique bichromatic opposite pair in the ith copy of K_4. Consequently, the b_i-pair and the c_i-pair are both monochromatic and receive different colors.

In the first copy, the b_1-pair has color 1 or 2. Suppose first that it has color 1. Then the c_1-pair has color 2. Since $\{c_3^+, b_1^-\}$ is a class of M_2, the c_3-pair must have color 3. The b_2-pair cannot have color 2, since $\{c_1^+, b_2^-\}$ is a class of M_2, and it cannot have color 3, since $\{b_2^+, c_3^-\}$ is a class of M_2. This is impossible, as the only available colors in the second copy are 2 and 3.

Suppose instead that the b_1-pair has color 2. Then the c_1-pair has color 1. Since $\{b_3^+, c_1^-\}$ is a class of M_2, the b_3-pair must have color 3. The c_2-pair cannot have color 2, since $\{b_1^+, c_2^-\}$ is a class of M_2, and it cannot have color 3, since $\{c_2^+, b_3^-\}$ is a class of M_2. This is again impossible. Hence $\chi_\ell(M_1, M_2) \ge 3$.

The above argument already shows that $\chi_\ell(M_1, M_2) > \chi(M_1, M_2)$. For completeness, we show that the list chromatic number is exactly 3. The graphic matroid M_1 is 2-colorable. By the partition-matroid reduction theorem of Bérczi, Schwarcz, and Yamaguchi [4, Theorem 1.6], there is a unit-capacity partition matroid N_1 on the same ground set such that every N_1-independent set is M_1-independent and $\chi(N_1) \le 2\chi(M_1) - 1 = 3$. The matroid M_2 is already a unit-capacity partition matroid and is 2-colorable. The common independent sets of N_1 and M_2 are the matchings of a bipartite multigraph whose two vertex classes correspond to the partition classes of N_1 and M_2. Since every class of N_1 has size at most 3 and every class of M_2 has size at most 2, this multigraph has maximum degree at most 3. By Galvin's theorem [9], it has a proper list edge-coloring from arbitrary lists of size 3. Each color class is independent in both N_1 and M_2, and hence also in both M_1 and M_2. Therefore $\chi_\ell(M_1, M_2) \le 3$. □

2.2 Two strongly base-orderable matroids

The example above does not settle the strongly base-orderable case. Indeed, M_1 has $M(K_4)$ as a minor, while $M(K_4)$ is not base-orderable [6]; since strongly base-orderable matroids are minor-closed and in particular base-orderable, M_1 is not strongly base-orderable. When two colors suffice, the partition-matroid reduction viewpoint gives a positive result.

Proposition 2.2. *Let $M = (E, \mathcal{I})$ be a strongly base-orderable matroid, and suppose that $E = I \cup J$ for two disjoint independent sets I and J. Then M has a reduction to a unit-capacity partition matroid whose classes have size at most two.*

Proof. Extend I to a basis $B_1 = I \cup C$ with $C \subseteq J$, and extend J to a basis $B_2 = J \cup A$ with $A \subseteq I$. Then $B_1 \cap B_2 = A \cup C$, $B_1 \setminus B_2 = I \setminus A$, and $B_2 \setminus B_1 = J \setminus C$.

Apply strong base orderability in the contraction $M/(A \cup C)$ to the disjoint bases $I \setminus A$ and $J \setminus C$. There is a bijection $\varphi\colon I \setminus A \to J \setminus C$ such that $((I \setminus A) \setminus X) \cup \varphi(X)$ is a basis of $M/(A \cup C)$ for every $X \subseteq I \setminus A$. Let N be the unit-capacity partition matroid whose classes are the pairs $\{x, \varphi(x)\}$ for $x \in I \setminus A$, together with the singletons $\{e\}$ for $e \in A \cup C$.

To see that N is a reduction of M, let T be independent in N, and let X be the set of those $x \in I \setminus A$ for which $\varphi(x) \in T$. Then $T \subseteq ((I \setminus A) \setminus X) \cup \varphi(X) \cup A \cup C$. The first two terms form a basis of $M/(A \cup C)$, while $A \cup C$ is independent in M, so the set on the right is a basis of M. Hence T is independent in M. □

Corollary 2.3. *Let M_1 and M_2 be strongly base-orderable matroids on a common ground set. If $\chi(M_1) \leq 2$ and $\chi(M_2) \leq 2$, then $\chi_\ell(M_1, M_2) \leq 2$. In particular, if $\chi(M_1, M_2) = 2$, then $\chi_\ell(M_1, M_2) = 2$.*

Proof. Applying Proposition 2.2 to each matroid gives unit-capacity partition matroids N_1 and N_2 such that every N_j-independent set is M_j-independent and every partition class of N_j has size at most two. The common independent sets of N_1 and N_2 are the matchings of a bipartite multigraph of maximum degree at most two. By Galvin's theorem [9], this multigraph has a proper list edge-coloring from arbitrary lists of size two. The resulting color classes are independent in both M_1 and M_2, so $\chi_\ell(M_1, M_2) \leq 2$. The equivalent formulation follows from the theorem of Davies and McDiarmid mentioned in Section 1.3. □

3 Concluding questions

The equality $\chi_\ell = \chi$ fails for two matroids, so it is natural to ask how large the additive gap can be. Aharoni, Berger, Guo, and Kotlar [2] proved $\chi_\ell(M_1, M_2) \leq 2\chi(M_1, M_2)$ for every pair of matroids, while Theorem 1.1 shows that an additive gap of one is possible.

Question 3.1. *Is there a constant C such that $\chi_\ell(M_1, M_2) \leq \chi(M_1, M_2) + C$ for every pair of matroids M_1, M_2? In particular, does $\chi_\ell(M_1, M_2) \leq \chi(M_1, M_2) + 1$ always hold?*

When one matroid is graphic and the other is a partition matroid, the reduction theorem of Bérczi, Schwarcz, and Yamaguchi [4, Theorem 1.6] and Galvin's theorem give

$$\chi_\ell(M_1, M_2) \leq \max\{2\chi(M_1) - 1, \chi(M_2)\} \leq 2\chi(M_1, M_2) - 1.$$

For $\chi(M_1, M_2) = 2$, this bound equals 3 and is attained by the example in Theorem 1.1. Thus the bound is tight when the common chromatic number is two; for $\chi(M_1, M_2) \geq 3$, it remains open whether it can be improved.

The strongly base-orderable setting remains open beyond two colors. Davies and McDiarmid's theorem determines the common chromatic number in this class [7], and Corollary 2.3 gives the list analogue when that number is two. Király and Pap [11] proved equality for the subclass of transversal matroids.

Question 3.2. *Does $\chi_\ell(M_1, M_2) = \chi(M_1, M_2)$ hold whenever M_1 and M_2 are strongly base-orderable?*

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